

Preliminary Proposal

Project Title: AI Forensic Triage Tool: Predicting Shooting Incidents in Boston to Prioritize Crime Lab Resources

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1. Introduction

Forensic crime laboratories and police departments process thousands of incidents annually under tight resource constraints. Ballistics analysis, DNA testing, and firearms-related evidence are time-intensive. However, not every reported crime involves a shooting. This project develops an AI Forensic Triage Tool that predicts shooting probability in real time using public Boston data, allowing labs to prioritize high-impact cases and improve public safety outcomes.

2. Problem Statement

Current evidence-processing workflows treat all incidents uniformly, creating backlogs and delaying justice in shooting-related cases. With over 300,000 incidents in the Boston dataset since 2015, manual triage is inefficient. A predictive model using incident features and neighborhood demographics can flag high-probability shooting cases for faster forensic analysis.

3. Data Source and Description

- Boston Police Crime Incident Reports (2023–present): ~150k rows CSV from <https://data.boston.gov/dataset/crime-incident-reports-august-2015-to-date-source-new-system>. Key columns: SHOOTING (binary target), OCCURRED_ON_DATE, DISTRICT, Lat/Long, OFFENSE_CODE_GROUP.
- U.S. Census ACS 2020–2024: Boston neighborhoods/tracts (median income, poverty rate, education, race, housing density) from data.census.gov. These will be merged by district to add socioeconomic context.

Both datasets are public, free, and require original cleaning — fulfilling capstone requirements.

4. Research Question and Hypothesis

Research Question: Can incident features (time of day, location, district, offense proxies) and neighborhood demographics accurately predict whether a reported crime will involve a shooting?

Hypothesis: Nighttime incidents in higher-poverty districts will show $\geq 35\%$ higher predicted probability of involving a shooting.

5. Variable Justification and Intersections

- Time features (hour_sin/cos, is_night, is_weekend) capture known violent-crime patterns.
- Location (DISTRICT + Lat/Long) reflects Boston's geographic disparities.
- Offense proxies (violent flag) serve as strong predictors.
- Census demographics (poverty, income, education, race) add the social determinants layer. Key interactions: night × high-poverty (expected top SHAP feature).

6. Methodology

6.1 Data Cleaning & Preprocessing: Parse dates, handle missing values, filter invalid coordinates, create binary SHOOTING target.

6.2 Feature Engineering: Circular time encoding, violent proxy, district dummies, census merge, SMOTE for class imbalance.

6.3 Exploratory Data Analysis: Shooting rates by district/hour, correlation matrices.

6.4 Hypothesis Testing: Chi-square tests and preliminary logistic regression.

6.5 Predictive Modeling: XGBoost / Random Forest classifier. Evaluation: Precision-Recall AUC, F1-score. SHAP explainability.

7. Planned Visualization

District heatmaps, time-of-day line plots, SHAP summary and dependence plots, and a final Tableau dashboard for stakeholder presentation.

8. Expected Key Insights

Nighttime and poverty will emerge as dominant predictors. The model is expected to achieve >0.80 PR-AUC with clear, actionable SHAP explanations for forensic triage.

9. Target Audience/Stakeholder

Massachusetts State Police Crime Laboratory and Boston Police Department — direct users for real-world case prioritization and resource allocation.

10. Conclusion

This Boston-specific, public-data project delivers a practical AI tool with immediate public-safety value while meeting all capstone requirements.

References

1. Boston Police Department. (n.d.). Crime Incident Reports (August 2015 to Date). City of Boston Open Data Portal. <https://data.boston.gov/dataset/crime-incident-reports-august-2015-to-date-source-new-system>
2. U.S. Census Bureau. (2020–2024). American Community Survey 5-Year Estimates. <https://data.census.gov>
3. Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. <https://doi.org/10.1145/2939672.2939785>
4. Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. Advances in Neural Information Processing Systems 30. <https://arxiv.org/abs/1705.07874>
5. Heller, S. B., et al. (2017). Thinking, fast and slow? Some field experiments to reduce crime and dropout in Chicago. The Quarterly Journal of Economics. <https://doi.org/10.1093/qje/qjw033>
6. Weisburd, D., et al. (2016). Place matters: Criminology for the twenty-first century. Cambridge University Press. (Chapter on geographic concentration of shootings in U.S. cities)
7. Braga, A. A., & Weisburd, D. (2012). The effects of focused deterrence strategies on crime: A systematic review and meta-analysis of the empirical evidence. Journal of Research in Crime and Delinquency. <https://doi.org/10.1177/0022427811419368>
8. National Institute of Justice. (2021). Predictive Policing: The Role of Crime Forecasting in Law Enforcement. <https://nij.ojp.gov/topics/articles/predictive-policing-role-crime-forecasting-law-enforcement>